P P SAVANI UNIVERSITY

Fifth Semester of B. Sc. IT Examination

November 2024

SSCS3021 Blockchain Technology

20.11.2024, Wednesday **Time: 09:00 a.m. To 11:30 a.m.** **Maximum Marks: 60**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

			CO	BTL	PO
Q - 1	MCQ/Short Question/Fill in the Blanks (Any Five)	[05]			
(i)	What is consortium blockchain?		01	01	01
(ii)	List the components of a business network.		02	04	03
(iii)	What is Hyperledger?		01	04	02
(iv)	What is the use of IBM world wire?		01	02	01
(v)	What is the role of committer?		02	02	03
(vi)	Enlist the principles of blockchain.		02	01	03
(vii)	Which are the types of assets in context of blockchain network?		03	04	02
Q - 2 (a)	Explain Private blockchain with their advantages and disadvantages.	[05]	4	4	2
Q - 2 (b)	Why we need blockchain solution over the traditional business network?	[05]	1	2	5
	OR				
Q - 2 (a)	Explain working of Blockchain with diagram.	[05]	1	2	1
Q - 2 (b)	Explain Proof of work with suitable diagram.	[05]	4	4	6
Q - 3 (a)	Explain the role of IBM food trust in improving supply chain transparency.	[05]	3	2	3
Q - 3 (b)	Explain Hyperledger fabric with suitable diagram.	[05]	2	1	8
	OR				
Q - 3 (a)	Explain the applications of blockchain in specific industry.	[05]	4	1	2
Q - 3 (b)	Define Merkle Tree with odd no. of transaction and its use with diagram.	[05]	3	2	2
Q - 4	Attempt any one.	[05]			
(i)	Enlist attack related to blockchain and explain any one of the attack in detail.		1	5	1
(ii)	Is it possible to rewrite the transaction once it is written in a block? Explain in detail.		2	3	2

SECTION – II

Q - 1	MCQ/Short Question/Fill in the Blanks (Any Five)	[05]			
(i)	Full form of Zk-SNARK.		2	2	3
(ii)	Full form of PoET.		4	4	3
(iii)	Hyperledger is a Blockchain. True/False?		1	1	1
(iv)	The Configtxgen tool is used to create which configuration artifacts?		4	2	1
(v)	Which is a mechanism that allows tokens to be securely transferred from one blockchain to another blockchain?		2	4	8
(vi)	Identity is anonymous in public Blockchain. True /False?		3	5	4
(vii)	Users treated equally in private blockchain. True / False?				
Q - 2 (a)	What are the difference between traditional and distributed system?	[05]	3	3	4
Q - 2 (b)	Explain Hyperledger composer along with its benefits	[05]	1	1	4

OR

Q - 2 (a)	Discuss the purpose and working of Proof-of-Work (PoW) Consensus.	[05]	1	2	2
Q - 2 (b)	Explain Private Blockchain network. Also compare it with Consortium Blockchain.	[05]	2	1	8
Q - 3 (a)	Draw and explain Hyperledger Greenhouse.	[05]	1	1	1
Q - 3 (b)	Explain actors in Blockchain Environment.	[05]	3	2	1

OR

Q - 3 (a)	Explain the core components of Blockchain architecture.	[05]	3	2	4
Q - 3 (b)	Compare at least three different type of real life blockchain based applications.	[05]	2	1	8
Q - 4	Attempt any one/two.	[05]			
(i)	Discuss Blockchain fabric development.		3	2	4
(ii)	What is consensus Mechanism? Explain any one in depth.		2	1	3

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CO : Course Outcome Number BTL : Blooms Taxonomy Level

Level of Bloom's Revised Taxonomy in Assessment

1: Remember	2: Understand	3: Apply
4: Analyze	5: Evaluate	6: Create